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FOC

克拉克逆变换

克拉克

 $M =$

$$\frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$\frac{2}{3} T_{3S/2S}$$

①

$$T_{2S/3S} = \frac{3}{2} T_{3S/2S}^{-1}$$

$$M^{-1} = \frac{1}{|M|} M^*$$

(求逆矩阵)

$$|M| = 1 \cdot \begin{vmatrix} \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} \end{vmatrix} - (-\frac{1}{2}) \begin{vmatrix} 0 & -\frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} \end{vmatrix} + (-\frac{1}{2}) \begin{vmatrix} 0 & \frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} \end{vmatrix}$$

$$= 1 \cdot \frac{\sqrt{3}}{2} + \frac{1}{2} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{\sqrt{3}}{4}$$

$$= \frac{3\sqrt{3}}{4}$$

$$M^* \rightarrow A_{ij} = (-1)^{i+j} M_{ji}$$

$$A_{11} = (-1)^2 \begin{vmatrix} \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} \end{vmatrix} = \frac{\sqrt{3}}{2}$$

$$A_{12} = (-1)^3 \begin{vmatrix} 0 & -\frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} \end{vmatrix} = \frac{\sqrt{3}}{4}$$

$$A_{13} = (-1)^4 \begin{vmatrix} 0 & \frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} \end{vmatrix} = \frac{\sqrt{3}}{4}$$

$$A_{21} = (-1)^3 \begin{vmatrix} -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{vmatrix} = 0$$

$$A_{22} = (-1)^4 \begin{vmatrix} 1 & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{vmatrix} = \frac{3}{4}$$

$$A_{23} = (-1)^5 \begin{vmatrix} 1 & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{vmatrix} = -\frac{3}{4}$$

$$A_{31} = (-1)^4 \begin{vmatrix} -\frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ 0 & \frac{\sqrt{3}}{2} \end{vmatrix} = \frac{\sqrt{3}}{2}$$

$$A_{32} = (-1)^5 \begin{vmatrix} 0 & \frac{\sqrt{3}}{2} \\ 0 & \frac{\sqrt{3}}{2} \end{vmatrix} = -\frac{\sqrt{3}}{2}$$

$$A_{33} = (-1)^6 \begin{vmatrix} 0 & -\frac{\sqrt{3}}{2} \\ 0 & \frac{\sqrt{3}}{2} \end{vmatrix} = \frac{\sqrt{3}}{2}$$

代数余子阵

$$\Rightarrow \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{4} & \frac{\sqrt{3}}{4} \\ 0 & \frac{3}{4} & -\frac{3}{4} \\ \frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

 M^* 为代数余子阵转置



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$$M^{-1} = \frac{1}{|M|} M^*$$

$$|M| = \frac{2\sqrt{3}}{4}, \quad |M| = \frac{4}{3\sqrt{3}}$$

$$M^{-1} = \frac{4}{3\sqrt{3}} M^*$$

$$\therefore M^{-1} = \begin{bmatrix} \frac{1}{3} & 0 & \frac{2}{3} \\ -\frac{1}{3} & \frac{\sqrt{3}}{3} & \frac{2}{3} \\ \frac{1}{3} & -\frac{\sqrt{3}}{3} & \frac{2}{3} \end{bmatrix}$$

$$T_{x/y} = \frac{3}{2} M^{-1}$$

$$= \begin{bmatrix} 1 & 0 & 1 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} & 1 \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} & 1 \end{bmatrix}$$

matlab坐标变换

$$f\alpha' = -\beta \Rightarrow$$

$$f\beta' = f\alpha$$

$$\Rightarrow \frac{2}{3} \begin{bmatrix} 0 & -\frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{2} \\ 1 & -\frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$



CS 扫描全能王

3亿人都在用的扫描App